

Windows Server vs Ubuntu Server vs Rocky Linux

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Proprietary commercial license; evaluation editions are available.	Open-source; Ubuntu Server is available under open-source licensing with optional commercial support.	Open-source enterprise Linux distribution with community and commercial ecosystem options.
User Interface	GUI is commonly available and administration can also be done through PowerShell/Server Core.	Primarily command-line on a server installation; GUI can be added when appropriate.	Primarily command-line on server installations; graphical tools can be added when needed.
Package Management	Uses Windows Update, MSI and Windows package/management tools.	Uses APT and Debian packages.	Uses DNF/RPM and the RHEL-compatible package ecosystem.
Security	Active Directory, Group Policy, Defender and Windows security tooling.	Strong Linux permissions, AppArmor, firewall tools and open-source security tooling.	SELinux, firewalld and enterprise-oriented security practices.
Performance	Strong for Microsoft workloads and integrated Windows services; resource use can be higher depending on roles.	Efficient and flexible for web, file, database, container and cloud workloads.	Efficient and stable for enterprise Linux workloads, especially RHEL-compatible environments.
Typical Enterprise Use Cases	Active Directory, Windows applications, Microsoft SQL Server, IIS, file/print services.	Web servers, cloud workloads, containers, development platforms, databases and general Linux infrastructure.	Enterprise servers, hosting, virtualization, databases and environments needing RHEL compatibility.
Advantages	Excellent Microsoft ecosystem integration, centralized management and broad enterprise tooling.	Free/open-source, large community, broad software availability and strong cloud support.	Enterprise-focused stability, RHEL compatibility, strong security and predictable lifecycle.
Disadvantages	Licensing costs and greater dependence on the Microsoft ecosystem.	Some enterprise products are less integrated than in Windows environments; administration is often CLI-focused.	Smaller desktop ecosystem and some software may target RHEL directly rather than Rocky specifically.

Discussion

Windows Server, Ubuntu Server, and Rocky Linux can all serve enterprise infrastructure roles, but they are optimized for different environments. Windows Server is especially strong where Microsoft technologies such as Active Directory, Group Policy, IIS, Microsoft SQL Server, and PowerShell are central. Its main trade-off is licensing and dependence on the Microsoft ecosystem.

Ubuntu Server is a flexible open-source platform commonly used for web hosting, cloud infrastructure, containers, development, databases, and general Linux server administration. Its APT package management, large community, and broad cloud support make it a practical choice for many organizations.

Rocky Linux is designed for enterprise environments that value compatibility with the Red Hat Enterprise Linux ecosystem. It provides a stable, security-focused Linux platform with tools such as SELinux and DNF. It is a strong option when an organization needs RHEL-compatible behavior without using a proprietary RHEL subscription for the operating system itself.

The best choice depends on workload, existing skills, security requirements, software compatibility, support model, and budget. A mixed environment is also common, with Windows Server used for Microsoft-specific services and Linux servers used for web, cloud, database, container, and application workloads.